Electrical Fuse Schedule (Implementation - Lynx Topology)

As-of date: `2026-03-20`

Purpose: define each required fuse by circuit, protected conductor/device, holder/housing method, physical placement, and linked wire-gauge assumptions for the approved Phase 1 Lynx architecture with a battery-backed 12V bus and dedicated 48V secondary alternator branch.

Related docs:

- Canonical electrical/system baseline: `docs/core/SYSTEMS.md`
- Implementation topology and conductor map: `docs/implementation/ELECTRICAL_overview_diagram.md`
- Decisions/open items tracker: `docs/core/TRACKING.md`
- BOM source of truth: `bom/bom_estimated_items.csv`

Design Basis

- Topology: `Victron Lynx Distributor M10` (`LYN060102010`) with `4` fused `48V` branches.
- Battery bank assumption: `3x 48V 100Ah` batteries in parallel (`3` separate battery-positive conductors leaving batteries).
- 12V distribution assumption: 12V fuse block used as the shared junction device (main `+` stud = source combine, integrated negative bus/main `-` = return), fed by Orion-Tr Smart `48/12-30` charger and a `12V 100Ah LiFePO4` buffer battery branch.
- Parallel-bank safety rule: use one Class T fuse per battery-positive conductor leaving the battery.
- Branch devices on Lynx:

1. MultiPlus-II `48/3000`
2. SmartSolar `150/45`
3. Dedicated `48V` secondary alternator branch (Mechman/WS500 path)
4. Orion-Tr `48/12-30` input

- Alternator migration lock for this pass:

1. `F-04` is re-baselined to `150A` MEGA (`58V/80V`) in Lynx Slot 3.
2. Legacy Sterling engine-bay input fuse path (`F-08`) is retired from active architecture.
3. WS500 support fuses are included as explicit install items (`F-12/F-13/F-14`).
4. Ford `Upfitter Switch #3` is locked as the manual `WS500` enable source through `F-15`.

Alternator Branch Fuse + Wire Finalization (`2026-03-19`)

Assumptions for this pass:

1. One-way alternator run length: `20 ft`.
2. Dedicated positive and dedicated negative runs (loop length `40 ft`).
3. Charging target voltage basis: `58.4V`.
4. Voltage-drop planning target: about `<=2%` under expected charging current.

Resistance and drop screen (`V_drop = I * (2 * L * R_per_ft)`):

Gauge	`150A` drop	`%` @ `58.4V`	`200A` drop	`%` @ `58.4V`	Decision note
`4 AWG`	`1.49V`	`2.55%`	`1.99V`	`3.40%`	Reject for this run length/current class
`2 AWG`	`0.94V`	`1.61%`	`1.25V`	`2.14%`	Acceptable for `~150A` class
`1/0 AWG`	`0.59V`	`1.01%`	`0.79V`	`1.35%`	Strong margin
`2/0 AWG`	`0.47V`	`0.80%`	`0.62V`	`1.07%`	Best margin; inventory already on hand

Lock for this build pass:

- Keep the planned alternator branch fuse at `150A` (`F-04`, Lynx Slot 3).
- Reuse existing uncut `2/0` inventory for alternator `+` and dedicated negative run.
- Row `173` contingency purchase is removed from scope unless field measurement proves an actual shortfall.

Required Fuse Map (Start-to-Finish, With Housing)

Fuse ID	Circuit (source -> load)	Protected wire/device	Fuse type and voltage class	Amperage	Holder or housing method	Physical location	Planned conductor gauge
F-01A	Battery A + -> bank positive combine/disconnect input	Battery A positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment, within ~`7"` of Battery A positive post	2/0 AWG
F-01B	Battery B + -> bank positive combine/disconnect input	Battery B positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment, within ~`7"` of Battery B positive post	2/0 AWG
F-01C	Battery C + -> bank positive combine/disconnect input	Battery C positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment, within ~`7"` of Battery C positive post	2/0 AWG
F-02	Lynx Slot 1 -> MultiPlus DC+	Main inverter positive feeder	MEGA, 58V or 80V	125A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 1	2/0 AWG planned (AWG 1 minimum on short run)
F-03	Lynx Slot 2 -> SmartSolar BAT+	MPPT battery-side positive feeder	MEGA, 58V or 80V	60A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 2	6 AWG
F-04	Lynx Slot 3 -> 48V alternator branch input (ALT+)	Alternator-to-house positive charge cable	MEGA, 58V or 80V	150A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 3	2/0 AWG (reuse lock)
F-05	Lynx Slot 4 -> 48V auxiliary feeder (Orion branch)	Aux feeder from Lynx to Orion-branch fuse point	MEGA, 58V or 80V	40A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 4	6 AWG
F-06	48V aux feeder -> Orion 48V input +	Orion input lead (device protection)	Inline MIDI family rated >=58VDC	30A (58V MIDI, lock)	Littelfuse 04980921GXM5 sealed inline MIDI holder on backplate	Electrical cabinet, mounted at source end of Orion input lead	6 AWG planned (8 AWG minimum per Orion cable table)
F-07	Orion 12V output + -> 12V fuse block main + stud	Main 12V feeder from Orion into shared source-combine point	MEGA, 58V or 80V	60A	Victron MEGA fuse holder (external, non-Lynx)	Electrical cabinet, within ~`7"of Orion12V` output stud	6 AWG planned (8 AWG minimum per Orion cable table)

F-09A/B/C	PV string + leads -> MPPT PV combiner	Each solar string positive conductor and reverse-current path	gPV string fuse (>=150VDC)	15A each (provisional)	10x38 touch-safe PV fuse holders in weatherproof combiner enclosure	Roof-entry combiner near gland/pass-through	10 AWG PV wire
F-10	12V fuse block branch circuits -> each 12V load	Individual 12V branch conductors and load circuits	ATO/ATC blade fuses (32V class)	Per-circuit	Integrated sockets in marine 12V fuse block	12V fuse block in electrical cabinet	Per branch
F-11	12V buffer battery + -> 12V fuse block main + stud via SW-12V-BATT	Buffer battery source cable and downstream junction fault exposure	Inline MIDI/AMI/ANL family rated >=32VDC	100A class baseline	Sealed inline holder mounted close to battery positive	Within ~`7"` of 12V buffer battery positive post	4 AWG planned
F-12	WS500 regulator power lead	WS500 power feed lead	Inline ATC/ATO (32V class acceptable on 12V-origin lead)	10A baseline (15A allowed if required by alternator case)	Sealed inline holder (WS500 kit)	Near source end of WS500 power lead	Harness lead
F-13	WS500 battery positive sense lead	WS500 battery sense lead	Fuse/holder rated for actual 48V bank maximum voltage unless WS500-supplied harness documentation proves otherwise	3A	Holder per WS500 manual/kit, voltage class verified before install	Near battery-positive sense takeoff	Harness lead
F-14	WS500 current-sense lead protection (when required by selected shunt layout)	WS500 current-sense wiring	Voltage class/reference to be confirmed from WS500 documentation before hardware lock	5A	Holder per WS500 manual/kit if required	Near shunt/sense tap per WS500 guidance	Harness lead
F-15	Ford Upfitter #3 -> WS500 brown ignition/enable wire	Low-current regulator enable/control wire	Inline ATC/ATO (32V class, local wire protection)	3A	Sealed inline holder near the Ford upfitter blunt-cut source / splice handoff	Engine bay or control-wire handoff point before small-gauge run to WS500	16 AWG TXL/GXL
OEM-SHUNT	Lynx positive tap -> SmartShunt positive sense/power lead	SmartShunt electronics lead	Victron OEM inline low-current fuse (factory harness)	OEM value	Integrated inline holder in supplied harness	Electrical cabinet near Lynx positive tap	OEM harness lead

Retired Fuse IDs (Migration Cleanup)

Retired ID	Previous path	Current status
F-08	Starter battery + -> Sterling BB1248120 input + (150A engine-bay fuse)	Retired with Sterling alternator-charge path. Hardware can remain as surplus/12V stock only.

Spare Fuse Inventory (Updated)

Fuse type	Installed qty	Spare qty to carry	Notes
Class T 200A (provisional installed)	3	1 on hand	Owner confirmed 3 holders and 4 slow-blow Class T fuses total (3 installed + 1 spare); add more only if the one-spare-per-installed policy is later desired
MEGA 150A (58V/80V)	1	2	Alternator branch (F-04) installed + spare set (row 170)
MEGA 125A (58V/80V)	1	4	MultiPlus branch
MEGA 60A (58V/80V)	2	4	Shared pool for MPPT (F-03) + Orion output (F-07)
MEGA 40A (58V/80V)	1	3	Orion feeder (F-05) plus spare set
Orion input fuse (30A MIDI, 58V)	1	3	BOM row 133 lock is x4 total (1 installed + 3 spares)
12V buffer battery main fuse (100A class)	1	3	Spare pack basis is BOM row 105
WS500 low-current fuses	3 active positions	2 each used value	Carry 10A, 15A, 5A, and 3A spares
WS500 ignition/enable fuse (F-15)	1 active position	2	Carry spare 3A mini/ATO fuse and one spare sealed holder if using a potted inline kit
PV string fuse 15A gPV	3	3	One spare per string
SmartShunt OEM harness fuse	1	1	Keep OEM-equivalent spare if field-replaceable
ATO/ATC branch fuses	variable	2 each used value	Keep mixed kit onboard

BOM Row Mapping

Fuse scope	BOM row(s)
Main battery Class T protection (F-01A/F-01B/F-01C) + Class T spares	bom/bom_estimated_items.csv row 7
Lynx branch MEGA fuses (F-02 to F-05) installed + spare set	bom/bom_estimated_items.csv rows 10 and 170
Orion installed fuse-holder hardware (F-06, F-07)	bom/bom_estimated_items.csv row 11
Orion input fuses (F-06)	bom/bom_estimated_items.csv row 133
WS500 low-current fuse/holder kit (F-12, F-13, F-14)	bom/bom_estimated_items.csv row 171
WS500 Upfitter #3 enable/control path (F-15)	bom/bom_estimated_items.csv row 176
12V buffer battery (B12)	bom/bom_estimated_items.csv row 21
12V buffer battery main fuse + holder (F-11)	bom/bom_estimated_items.csv row 125
12V battery disconnect (SW-12V-BATT)	bom/bom_estimated_items.csv row 124
12V branch panel and blade fuses (F-10)	bom/bom_estimated_items.csv row 16

SmartShunt OEM fused lead (`OEM-SHUNT`)	`bom/bom_estimated_items.csv` row `23` (included with SmartShunt kit)
PV string fuses + holder (`F-09A/B/C`) and spares	`bom/bom_estimated_items.csv` row `106`

Assumptions and Open Items

1. Wire sizing above assumes copper conductors and enclosed vehicle routing.
2. `F-09A/B/C` and the old `3S3P` fuse count are modeling placeholders only; final PV fusing waits until solar modules/stringing are selected after shore and alternator charging are working.
3. Confirm Mechman kit content/polarity (`PH` / `NH`) before physically returning Sterling hardware.
4. `SW-12V-BATT` remains manual-only in Phase 1; no automatic LVD behavior is assumed.
5. Final lock for `F-11` still requires explicit 12V buffer battery/BMS continuous discharge-current confirmation.
6. `F-15` exists to protect the smaller-gauge control wire between Ford `Upfitter #3` and the WS500 brown ignition/enable wire; the factory `25A` upfitter circuit protection is not the final wire-protection device for that branch.